{ "nbformat": 4, "nbformat_minor": 0, "metadata": { "colab": { "name": "Quadrature Time Simulation", "provenance": [] }, "kernelspec": { "name": "python3", "display_name": "Python 3" }, "language_info": { "name": "python" } }, "cells": [{ "cell_type": "markdown", "metadata": {}, "source": ["# The Hilbert-Space Quadrature Conjecture: Interactive Colab Notebook\n", "## State-Generated Quadrature Time in \mathbb{C}^N \otimes \mathbb{C}^2\n", "\n", "This notebook implements the finite-dimensional discrete model for **Quadrature Time Theory**. It demonstrates:\n", "1. The covariant Pegg-Barnett clock phase states on the circle S^1.\n", "2. Exact bipartite conditioning of a system qubit relative to a quantum clock of spread \Delta n.\n", "3. Finite-clock Lindblad dephasing and convergence to smooth Schr\"odinger dynamics as N \to \infty."] }, { "cell_type": "code", "execution_count": null, "metadata": {}, "outputs": [], "source": ["# Import dependencies\n", "import numpy as np\n", "import matplotlib.pyplot as plt\n", "from ipywidgets import interact, IntSlider, FloatSlider\n", "\n", "print('Environment initialized successfully.')"] }, { "cell_type": "code", "execution_count": null, "metadata": {}, "outputs": [], "source": ["def exact_bipartite_quadrature(N=128, Delta_n=8.0, omega_ratio=1.0):\n", " """\n", " Computes exact conditional quadrature dynamics directly from a\n", " bipartite entangled state |Psi> in C^N (clock) x C^2 (system)\n", " using a covariant smeared clock POVM of energy spread Delta_n.\n", " """\n", " theta = 2.0 * np.pi * np.arange(N) / N\n", " n = np.arange(N)\n", " n0 = N // 2\n", "\n", " # 1. Clock Readout POVM envelope with energy spread Delta_n\n", " env_n = np.exp(-(n - n0)2 / (4.0 * max(Delta_n, 1e-6)2))\n", " env_n /= np.linalg.norm(env_n)\n", "\n", " # 2. Ideal conditional qubit history states\n", " psi_ideal_theta = np.zeros((N, 2), dtype=complex)\n", " psi_ideal_theta[:, 0] = 1.0 / np.sqrt(2.0)\n", " psi_ideal_theta[:, 1] = np.exp(-1j * omega_ratio * theta) / np.sqrt(2.0)\n", "\n", " # 3. Intrinsic Clock Pointer Quadrature Profile G(theta)\n", " pointer_amp = np.fft.ifft(env_n) * np.sqrt(N)\n", " G_profile = np.abs(pointer_amp)2\n", " G_profile /= G_profile.sum()\n", "\n", " # 4. Exact conditional density matrix via cyclic convolution on S^1\n", " rho_ideal = np.einsum("ki,kj->kij", psi_ideal_theta, np.conj(psi_ideal_theta))\n", " rho_finite = np.zeros_like(rho_ideal)\n", " for k_idx in range(N):\n", " weights = np.roll(G_profile, k_idx)\n", " rho_finite[k_idx] = np.einsum("l,lij->ij", weights, rho_ideal)\n", "\n", " # 5. Analytical Gaussian / Lindblad dephasing prediction\n", " sigma_theta = 1.0 / (2.0 * Delta_n)\n", " analytical_coh = 0.5 * np.exp(-0.5 * (omega_ratio * sigma_theta)2)\n", "\n", " return theta, G_profile, rho_ideal, rho_finite, sigma_theta, analytical_coh"] }, { "cell_type": "code", "execution_count": null, "metadata": {}, "outputs": [], "source": ["# Interactive Widget for Jupyter / Google Colab\n", "def interactive_simulation(N=128, Delta_n=8.0, omega_ratio=2.0):\n", " theta, G_prof, rho_id, rho_fin, sig_th, ana_coh = exact_bipartite_quadrature(\n", " N=N, Delta_n=Delta_n, omega_ratio=omega_ratio\n", ")\n", "\n", " fig, axs = plt.subplots(1, 2, figsize=(14, 5))\n", "\n", " # Panel 1: Clock Pointer Quadrature Profile on S^1\n", " axs[0].plot(theta, np.fft.fftshift(G_prof), "o-", markersize=3, color="#1f77b4")\n", " axs[0].set_xlabel(r"Clock Phase Angle \theta \in [0, 2\pi)", fontsize=11)\n", " axs[0].set_ylabel(r"Pointer Quadrature G_{\sigma_\theta}(\theta)", fontsize=11)\n", " axs[0].set_title(rf"Clock Quadrature Kernel (N={N}, \Delta n={Delta_n})", fontsize=12)\n", " axs[0].grid(True, alpha=0.3)\n", "\n", " # Panel 2: Qubit Coherence rho_01(theta)\n", " axs[1].plot(theta, np.real(rho_id[:, 0, 1]), "--", color="gray", alpha=0.7,\n", " label=r"Ideal \mathrm{{Re}}[\rho_{01}(\theta)] Guard")\n", " axs[1].plot(theta, np.real(rho_fin[:, 0, 1]), "-", color="#d62728", lw=2,\n", " label=r"Finite-Clock \mathrm{{Re}}[\rho_{01}^{\mathrm{obs}}(\theta)] Collective")\n", " axs[1].axhline(ana_coh, color="#2ca02c", ls=":", lw=1.8,\n", " label=rf"Analytical Envelope \vert{}\rho_{{01}}\vert{} = {ana_coh:.4f}")\n", " axs[1].axhline(-ana_coh, color="#2ca02c", ls=":", lw=1.8)\n", " axs[1].set_xlabel(r"Emergent Clock Phase \theta = \omega t", fontsize=11)\n", " axs[1].set_ylabel(r"Conditional Coherence \rho_{01}(\theta)", fontsize=11)\n", "

axs[1].set_title("Conditional Schr\"odinger Evolution & Clock Dephasing", fontsize=12)\n", " axs[1].set_ylim(-0.58, 0.58)\n", " axs[1].legend(loc="upper right", fontsize=9)\n", " axs[1].grid(True, alpha=0.3)\n", "\n", " plt.tight_layout()\n", " plt.show()\n", "\n", "interact(\n", " interactive_simulation,\n", " N=IntSlider(min=32, max=256, step=32, value=128),\n", " Delta_n=FloatSlider(min=2.0, max=32.0, step=1.0, value=8.0),\n", " omega_ratio=FloatSlider(min=0.5, max=5.0, step=0.5, value=2.0)\n", ");"] }] }